Class-Level Prototype Guided Multiscale Feature Learning for Remote Sensing Scene Classification With Limited Labels

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***Abstract*— Remote sensing scene classification (RSSC) is an open and challenging research topic in the remote sensing (RS) community. It aims to define semantic labels for RS scenes according to their contents. Recently, with the development of deep convolutional neural networks (DCNNs), the results of RSSC have been enhanced to a large extent. However, the cracking performance of these DCNN-based models depends on a large number of labeled data. Once the volume of the labeled data is decreased, their behavior would be weakened dramatically. In this article, we propose a new training algorithm that can work smoothly with a few labeled samples to address this limitation. Along with the introduced DCNN, the presented methods perform satisfactorily. In particular, we first construct a dual-branch network (DBNet) to mine the multiscale and multiangle information from RS scenes. Thus, the abundant land covers with diverse sizes, directions, and shapes can be captured simultaneously. Then, to train DBNet using scarce semantic labels, a class-level prototype guided learning (CPGL) algorithm is developed based on the meta-learning paradigm. Besides the usual episode training manner, a prototype refinement module (PRM) and a prototype discrimination module (PDM) are designed with the help of metric learning theory to ensure the effectiveness of our CPGL. The comprehensive experiments are conducted on four public RS scene datasets, and the encouraging results imply that our DBNet and CPGL can copy with RSSC tasks with small labeled data. Our source codes are available at https://github.com/TangXu-Group/Remote-Sensing-Images-Classification/tree/main/CPGL.**

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***Index Terms*— A few labeled samples, deep convolutional neural networks (DCNNs), remote sensing scene classification (RSSC).**

I. Introduction

With the development of sensor technology, a large number of remote sensing (RS) images with high resolution can be collected every day. These massive RS images bring sufficient useful land-cover/land-use information. However, their abundant, complex, and varied contents also bring many challenges in interpreting them. As a basic interpretation task, RS scene classification (RSSC) has received considerable attention for a long time as it is the first step for many RS image processing tasks, including land cover classification [1], change detection [2], object detection [3]–[5], RS image caption [6], [7], and content-based RS image retrieval [8]–[10]. RSSC aims at assigning semantic labels to RS scenes based on their contents [11]–[13], and it can be used in many practical applications, such as disaster detection [14] and geospatial object surveying and mapping [15].

In the last decades, many RSSC methods have been proposed [16]. From observing these approaches, it is easy to find that their performance largely depends on the RS image representation. In other words, visual features play vital roles in RSSC. In the beginning, low-level visual features are popular as they are simple in accomplishment and stable in performance. The common low-level descriptors are color, texture, and shape [17]–[19]. Then, mid-level visual features combined with low-level features according to some specific rules attract scholars' eyes. The prevalent mid-level descriptors are bag-of-visual-words (BoVW) [20], vectors of locally aggregated descriptors (VLAD) [21], and improved Fisher kernel (IFK) [22]. The mentioned low- and mid-level visual descriptors are hand-crafted. Along with these features, classifiers that are developed based on machine learning and statistical theories are adopted to accomplish the classification, including support vector machine (SVM) [23], random forest [24], and so on.

With the booming of deep learning [25], deep convolutional neural networks (DCNNs) [26]–[28] provide new opportunities for RSSC [29]. Due to the strong capacity of nonlinear fitting, the specific hierarchical structure, and abundant supervised prior knowledge, DCNNs can learn high-level

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visual features from RS images. These descriptors are rich in semantic knowledge that is beneficial to classification tasks. Besides, DCNN s integrate feature learning and classification into an end-to-end framework, which leads the classifier to follow the feature distributions well. Although many supervised DCNN-based methods [30]–[33] improve the performance of RSSC adequately, their shortcoming is distinct, i.e., they need a large number of labeled data to complete the training process. This is a rigorous demand since annotating RS images manually is hard or even impractical in some cases. To mitigate this problem, some scholars design data augmentation strategies [34], [35] to enlarge the scale of labeled data. Others develop the deep models under the semisupervised learning paradigm [36]–[38]. Although they are feasible, their performance cannot reach what we expect. The main reasons are: 1) data augmentation aims at generating more samples rather than enriching their diversity, which leads to that the trained models are always poor in generalization and 2) semisupervised methods should meet some assumptions [39] that limit their practicability.

Recently, to address the limited training data problem for classification tasks, few-shot learning (FSL) [40] enters the RS community. As a valuable solution for FSL, meta-learning [41], [42] uses the episode training (N-way-K-shot learning manner) to simulate the actual few-shot scenario and accomplishes the training process using a few samples with supervised information. By expanding the training units from a single sample to a set of tasks, meta-learning models first learn how to cover the data distribution under limited samples in the training stage and then generalize this learning ability to the testing set. In this way, both the performance and generalization of classification models can be guaranteed. Although many meta-learning approaches have been successfully developed for various FSL tasks [43], [44], it is worth noting that there is a practical gap between them and real applications. Particularly, due to the specific N-way-K-shot training manner, meta-learning models always need a set of tasks to complete the inference. In other words, it is intractable to use the trained meta-learning models to predict an unlabeled single sample directly. This is not able to meet the popular classification demands in practical scenarios.

After the above discussion, a question arises: is it possible to train a traditional DCNN with extremely scarce labels without overfitting? To this end, we propose a new learning method in this article. First, considering the complex contents within RS scenes (i.e., objects are diverse in type and scale), we develop a dual-branch network (DBNet) for RSSC, which focuses on exploring the multiscale and multiangle features from RS scenes. Second, inspired by meta-learning techniques, a specific training method is proposed for DBNet that aims to optimize its parameters using a little supervised information without overfitting problems. We record this training method class-level prototype guided learning (CPGL). CPGL consists of a prototype refinement module (PRM) and a prototype discrimination module (PDM). PRM adopts the metric learning theory [45] to impact intraclass samples, while PDM leverages the distance penalty strategy to separate interclass samples. Finally, our DBNet can be trained by a joint learning scheme with extremely deficient prior information. The trained DBNet can be used to predict a single RS scene directly.

Note that, apart from the meta-learning techniques, there are still some other solutions for the few-shot scenario, such as semisupervised learning [36] and transfer learning [46]. However, we insist on meta-learning because of the following reasons. First, the prototype-based meta-learning technique has inherent advantages in tackling the problem of big intraclass variance within RS scenes [47]. Second, meta-learning refers to a paradigm that can improve the other few-shot scenario methods [48]. Last but not least, in comparison to other few-shot solutions, meta-learning is more flexible since it aims to generalize and learn the learning process [49], which makes the obtained model adapted to different tasks. To sum up, we select meta-learning as the basis of the proposed CPGL algorithm in this article.

The main contributions of this article are summarized as follows.

1) To exploit the complex contents hidden in RS scenes, an end-to-end DBNet is proposed with the consideration of multiscale and multiangle properties. Due to the specific structure, this network can discover various land covers with diverse information.
2) To make DBNet work with a few labeled samples, a simple yet effective training method (CPGL) is proposed based on meta-learning techniques. To capture the intraclass and interclass relationships between RS scenes, PRM and PDM aim at generating representative and discriminative class-level prototypes in the embedding space. Therefore, our DBNet can be trained to distinguish scenes from different semantics with limited supervised information.
3) The comprehensive experiments are conducted on four public RS scene datasets. The competitive results demonstrate that our model and corresponding training method are helpful to RSSC tasks under the few sample scenarios.

The remaining sections are organized as follows. Related work is reviewed in Section II. Then, the proposed RSSC network and its training method are introduced in Section III. The evaluations of the proposed method are given in Section IV, Finally, the conclusions are summarized in Section V.

II. Related Work

In this section, we review the published DCNN-based RSSC methods. For clarity, we divide them into two groups, i.e., DCNN-based methods with sufficient and few labeled samples.

A. Methods With Sufficient Labeled Samples

In this group, we assume that there are many samples with supervised information for training deep models. It is well-known that the contents within RS scenes are complicated. The types and volume of land covers in an RS scene are diverse and enormous. Besides, the scales of objects in an RS scene are multiple. To accurately define the semantics for an RS scene, discriminative feature representation is needed for describing the complex contents from different hierarchies. Thus, scholars

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pay their attention to integrate the various features with different receptive fields [50]. For example, He *et al.* [51] developed a multilayer stacked covariance pooling to fuse features from different convolutional layers, which provides a natural and innovative way to exploit the complementary information. Lu *et al.* [52] introduced a progressive aggregation strategy to leverage the semantic label information to aggregate the intermediate features. The reported expressive classification results demonstrate its usefulness. With the help of the enhanced feature pyramid network, an RSSC model was proposed in [53], where both the multilevel features and deep semantics are used to distinguish different RS scenes.

Apart from enhancing the feature representation by the feature aggregation, researchers found that the capacity of visual descriptors can be improved if some key and important local information could be captured from assorted contents. Thus, the attention mechanism [54] is introduced in the RSSC community. Based on the long short-term memory (LSTM) model [55], a deep network with recurrent attention structure was proposed in [56] to produce an adaptive and dynamic attention mask matrix, which helps the network find the key regions hidden in RS scenes. To utilize the saliency regions deeply for RSSC, the context relationships between them can be supplemented by the global information. For instance, Guo *et al.* [57] presented a deep model to explore global and local semantics within RS scenes simultaneously, in which the fully connected layers are replaced by the global and local attention blocks. The classification results based on the fused features are competitive.

The works mentioned above focus on content representation. Nevertheless, the relationships between interclass and intraclass scenes are also essential to RSSC. To maximize intraclass similarity while minimizing interclass variation, Cheng *et al.* [58] imposed a metric learning regularization term during the feature learning using the Siamese network [59]. This operation enforces the RS scenes from the different classes, which can be far from each other in the feature space. Tang *et al.* [32] developed an attention consistent network (ACNet) to address RSSC. Besides exploring the local and global information from RS scenes, the network also considers the distances between interclass and intraclass scenes by a simple yet useful consistent block.

B. Methods With Few Labeled Samples

In limited samples scenarios, semisupervised learning is a popular solution, which aims at using both labeled and unlabeled data to accomplish the classification. For example, Zhang *et al.* [36] introduced the center loss [60] into a semisupervised network by using the unlabeled samples to update class centers. The positive results confirm its effectiveness. A pseudo label generation algorithm was proposed in [61] so that both labeled and unlabeled data can be used to train the classification model. Han *et al.* [62] proposed a semisupervised generative network based on a cotraining method. It can be used for annotating unlabeled RS scenes and extending labeled datasets. To effectively characterize RS scenes, an RSSC deep model was introduced in [63], which adopts the deep metric learning scheme and accomplishes classification under the semisupervised paradigm. Another solution for FSL is transfer learning [64], which focuses on transmitting the knowledge from source domains to target domains. Although the differences between source domains (natural images, such as ImageNet) and target domains (different types of RS images) are large, transfer learning still performs well in RSSC [46]. Therefore, it often acts as an auxiliary technique. For instance, the pretrained EfficientNet [65] and Res2Net [66] were fine-tuned in [67] and [68] for RSSC tasks. Different from directly using the pretrained DCNNs for feature learning, Liang *et al.* [69] propose a specific fine-tuning strategy that results in better DCNNs for RSSC.

In contrast to the training scheme used in the above methods, the models based on meta-learning take advantage of the episode-based training to accomplish RSSC tasks using a few labeled samples. Many meta-learning methods have been in recent years, such as prototypical networks [45], matching networks [43], and relation networks [44]. Based on those works, Li *et al.* [70] proposed an FSL model for RSSC, in which the training samples are organized in a task-driven way, and a metric space is learned to classify RS scenes from a series of tasks. Furthermore, a balance loss was developed to maximize the distances between scenes from different categories. Discriminative learning of adaptive match network (DLA-MatchNet) was presented in [71] under the framework of meta-learning. Rather than developing new metalearners or discovering transferable information, DLA-MatchNet emphasizes the representation and metric learning for RSSC. Both attention techniques and distance metric theory are utilized in the FSL framework. Integrated with knowledge distillation [72] and the episode learning manner, a ranking network [73] is developed for RSSC, which consists of feature learning and rank learning submodules. It forces the student network to learn the important rank of query features relative to support features determined by teacher networks and achieves encouraging results in few-shot RSSC.

III. METHODOLOGY

A. Preliminary Knowledge

To obtain effective and generalized deep models using a few labeled data, meta-learning techniques use a set of tasks instead of the samples to complete the training process. Under the framework of meta-learning, each task T consists of a few samples that are grouped into support and query sets $\{D_{\text{support}}, D_{\text{query}}\}$. They aim to estimate and update models' parameters, respectively. The collection of tasks $\mathcal{T}$ follows the distribution $p(\mathcal{T})$, i.e., $T \sim p(\mathcal{T})$.

Particularly, suppose that we have a labeled dataset that contains M classes with L samples per class. Note that L is very small, such as 5, 8, or 10. To construct a task in the meta-learning paradigm, we randomly construct the support set as $D_{\text{support}} = \{(x_i^s, y_i^s)\}_{i=1}^{N\times K}$ and the query set as $D_{\text{query}} = \{(x_i^q, y_i^q)\}_{i=1}^{N\times P}$, where N ($N < M$) denotes the class number, and K and P are the numbers of samples per class in the support and query sets and $K+P = L$. Then, the deep models can be trained with a set of N-way-K-shot task $\mathcal{T}$ iteratively,

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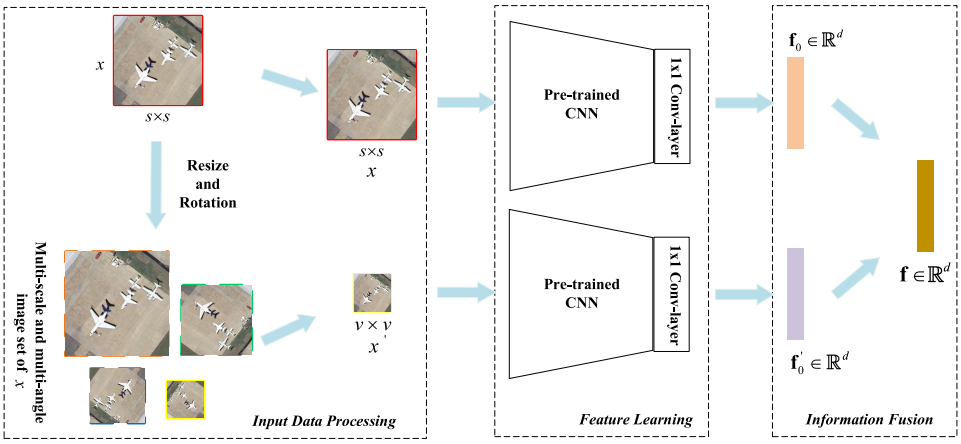


Fig. 1. Framework of our dual-branch feature learning network.

and the training process can be formulated as

$$\theta = \arg\min_{\theta} \sum_{\mathcal{T}} \mathcal{L}(f_\theta(T)) \tag{1}$$

where $\mathcal{L}(\cdot)$ denotes the loss function and $f_\theta(\cdot)$ indicates the nonlinear mapping of deep models. This manner discussed above is known as the episode training. In the inference stage, to use the trained metamodels to predict the unlabeled samples, they should also be constructed as different tasks.

B. Dual-Branch Feature Learning Network

Inspired by [74], we develop a deep model (DBNet) to capture the complex contents hidden in RS scenes, and its framework is exhibited in Fig. 1. It consists of an input data processing submodule, a feature learning submodule, and an information fusion submodule.

The input data processing submodule aims at highlighting the diverse scale and direction information within RS scenes by constructing various RS scene pairs. In detail, assume that there is an input RS scene x with the size of $u \times u$. We resize it to a set of new RS scenes. Their sizes are $v \times v$, where v is varied in a range of $v \in \{u \pm 10\tau, \tau = 1, 2, 3, 4, 5, 6\}$. Meanwhile, these resized scenes are rotated with 0°, 90°, 180°, and 270°, arbitrarily. Then, one of them is selected to construct the scene pairs (x, x'). To enhance the robustness of DBNet, x' is chosen randomly in each iteration. The feature learning submodule has a dual-branch structure, which focuses on capturing high-level features from the scene pairs x and x', respectively. Note that two convolutional neural networks (CNNs) are the same, but they do not share weights. In each branch, the CNN is established by a pretrained deep model and a 1×1 convolutional layer. Their function is to extract visual features from scenes and enhance the nonlinear ability of the network. After this submodule, x and x' are mapped into d-dimensional features $\mathbf{f}_0$ and $\mathbf{f}'_0$. Note that, in this article, we adopt ResNet-18 [28] as the pretrained deep model. To integrate the learned knowledge corresponding to different scales, the simple information fusion submodule is embedded at the top of DBNet. Through the average operation $(\mathbf{f}_0 + \mathbf{f}'_0)/2$, we can get the final visual feature $\mathbf{f} \in \mathbb{R}^d$ for the RS scene x.

In general, we can use abundant training data to train our DBNet with the cross-entropy loss function, which is defined as

$$\mathcal{L}_{\text{cls}} = -\frac{1}{|B|} \sum_{i}^{|B|} (y_i \log(\hat{y}_i)) \tag{2}$$

where y_i and $\hat{y}_i$ mean the ith sample's ground truth and prediction, and $|B|$ denotes the number of training images per batch. However, the problem that we met is that only a few labeled data are available. Consequently, we cannot use the traditional training method and the loss function to train our DBNet directly. A simple yet useful training strategy is developed to address this issue, and detailed information can be found in Section III-C.

C. Class-Level Prototype Guided Learning

The flowchart of the CPGL strategy is shown in Fig. 2. Suppose that we have a labeled RS scene set $\{x_1, \ldots, x_{ML}\}$ containing M classes with L scenes per class, where L is very small. To train our DBNet using these labeled data without overfitting and get the final d-dimensional high-level features $\mathbf{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_{ML}\} \in \mathbb{R}^{ML \times d}$, the proposed CPGL strategy formulates the training process in a meta way. Particularly, rather than the samples, a task set $D_{\text{train}} = \{(D^t_{\text{support}}, D^t_{\text{query}}) | t = 1, \ldots, I\}$ is used to train DBNet, which is constructed by the labeled RS scene set under the meta-learning paradigm. Here, I means the number of tasks. When a task $D^t_{\text{train}} = (D^t_{\text{support}}, D^t_{\text{query}})$ is passed through DBNet, its scene pairs $\{(x_1, x'_1), \ldots, (x_{NL}, x'_{NL})\}$ are mapped into a high-level feature matrix $\mathbf{F}^t = \{\mathbf{f}^t_1, \ldots, \mathbf{f}^t_{NL}\} \in \mathbb{R}^{NL \times d}$, where $\mathbf{f}^t_i$ implies the high-level feature of the ith scene in the task D^t_{train}, N denotes the class numbers in the task D^t_{train}, and d means the dimension of high-level features. To guarantee the discrimination of $\mathbf{F}^t$, apart from the cross-entropy loss function, we also develop PRM and PDM loss functions inspired by the prototypical network [45]. Note that two

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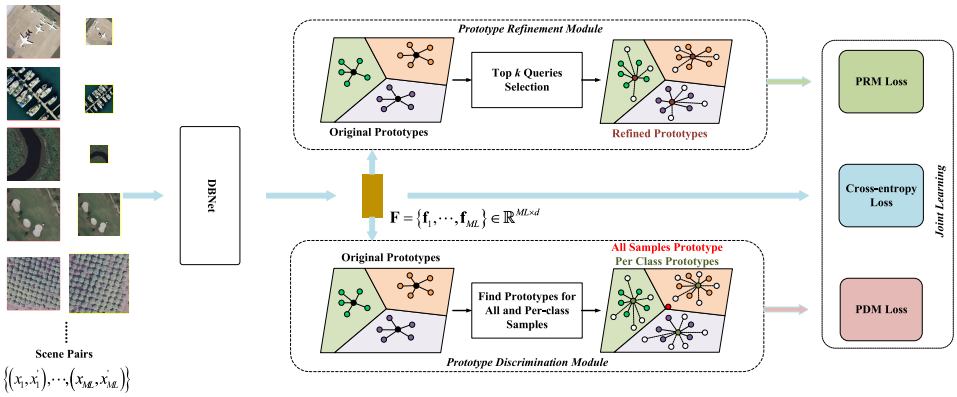


Fig. 2. Flowchart of the CPGL strategy.

new loss functions only work in the training stage. In the inference stage, they would be removed. By optimizing three loss functions jointly, our DBNet can be trained with a few numbers of labeled data. Now, we will discuss them in detail.

1) Prototype Refinement Module: Similar to prototypical networks, PRM aims to find helpful class prototypes for the subsequent inference. Different from prototypical networks, our PRM considers both support and query samples to generate class prototypes.

For each task, prototypical networks seem the mean value of mapped support samples within a class as the prototype, which is defined as

$$\mathbf{c}_i = \frac{1}{K} \sum_{\left(\left(x_j^s, y_j^s\right)\in S_i\right)} \mathbf{f}_j^s \tag{3}$$

where S_i denotes the set of support samples categorized in the ith class, K means the number of support samples within S_i, (x_j^s, y_j^s) denotes the jth support sample and its manual label in S_i, and $\mathbf{f}_j^s$ indicates the embedding of x_j^s. Then, for a query sample x_j^q, its prediction can be obtained according to the distances between it and different prototypes. This process can be formulated as

$$p\left(\hat{y}_j^q = i \middle| x_j^q\right) = \frac{\exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_i\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_k\right)\right)} \tag{4}$$

where $\mathbf{f}_j^q$ indicates the embedding of x_j^q and $d(\mathbf{f}_j^q, c_k)$ denotes the distance between $\mathbf{f}_j^q$ and c_k. In this article, we select the Euclidean distance unless otherwise noted.

Since the contents within RS scenes are complex, the prototypes only determined by the support samples are not representative enough to represent the accurate class distributions, limiting our proposed method's performance. To overcome this problem, PRM refines the prototype by introducing the query samples with high confidence. In particular, for the tth task $D_{\text{train}}^t = (D_{\text{support}}^t, D_{\text{query}}^t)$, its original prototypes can be obtained according to (4). Then, we count the distances between queries and prototypes, and sort them from small to large. We believe that the smaller the distance, the higher the confidence. Thus, top k^{top} queries would be selected to refine the prototypes. We record the selected queries $D_{sq}^t = \{(x_i^{sq}, y_i^{sq}), i = 1, \ldots, N \cdot k^{\text{top}} k^{\text{top}}\}$. Considering the support and selected simultaneously, the refined prototypes can be formulated as

$$\mathbf{c}_i^* = \frac{1}{K + k^{\text{top}}} \left(\sum_{\left(\left(x_j^s, y_j^s\right)\in S_i\right)} \mathbf{f}_j^s + \sum_{\left(\left(x_l^{sq}, y_l^{sq}\right)\in Q_i\right)} \mathbf{f}_l^{sq} \right) \tag{5}$$

where S_i and Q_i denote the sets of support samples and selected query samples categorized in the ith class, K and k^{top} are the number of support and selected query samples within S_i and Q_i, (x_j^s, y_j^s) implies the jth sample and its manual label in S_i, $\mathbf{f}_j^s$ indicates the embedding of x_j^s, (x_l^{sq}, y_l^{sq}) illustrates the lth sample and its manual label in Q_i, and $\mathbf{f}_l^{sq}$ means the embedding of x_l^{sq}. Based on the refined prototypes, the prediction of a query sample x_j^q can be defined as

$$p^*\left(\hat{y}_j^q = i \middle| x_j^q\right) = \frac{\exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_i^*\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(\mathbf{f}_j^q, \mathbf{c}_k^*\right)\right)} \tag{6}$$

where $\mathbf{f}_j^q$ indicates the embedding of x_j^q and $d(\mathbf{f}_j^q, c_k^*)$ denotes the Euclidean distance between $\mathbf{f}_j^q$ and c_k^*.

Finally, the original and refined prototypes are taken into account to define the PRM loss function [see (7)]. This operation is inspired by the residual connection in ResNet [28], which can reduce information loss and ensure the stability and performance of our PRM

$$\mathcal{L}_{\text{PRM}} = -\log\left(p^*\left(\hat{y}_j^q = i \middle| x_j^q\right)\right) - \log\left(p\left(\hat{y}_j^q = i \middle| x_j^q\right)\right). \tag{7}$$

2) Prototype Discrimination Module: PRM leverages extra information from interclass query samples to produce more representative corresponding prototypes, which results in a compacting intraclass feature space. However, the relationships between different class-labeled samples are not considered in PRM. Due to the complex contents within RS scenes, the diversity of intraclass samples is large, and the similarities of interclass samples are small. PRM has been proposed to alleviate the intraclass problem to some extent. To disperse and further compact the interclass and intraclass scenes, we develop PDM here. The goal of PDM is to push samples far away from the center and pull intraclass samples together. For a task $D_{\text{train}}^t = (D_{\text{support}}^t, D_{\text{query}}^t)$, its definition is

$$\mathcal{L}_{\text{PDM}} = \frac{1}{NL} \frac{\sum_{i=1}^{N} \|\mathbf{c}_i - \bar{\mathbf{c}}_i\|_2^2}{\sum_{i=1}^{N} \|\bar{\mathbf{c}}_i - \mathbf{c}\|_2^2} \tag{8}$$

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Algorithm 1 Specific Steps of CPGL for Training DBNet

Input:

A labeled data set that contains M classes with L samples per class, the maximum iterations $iterMax$.

Initialize DBNet with the pre-trained weights.
for $i = 1$ to $iterMax$ **do**
  Construct one training task D^t_{train},
  Generate RS scene pairs for the sample within D^t_{train} according to the scheme mentioned in Section III-B,
  Pass the obtained RS scene pairs through DBNet to get their features,
  Compute original prototypes $\mathbf{c}$ and refined counterpart $\mathbf{c}^*$ by Eq. 3 and Eq. 5,
  Calculate $\mathcal{L}_{cls}$, $\mathcal{L}_{PRM}$, and $\mathcal{L}_{PDM}$ by Eq. 10, Eq. 7, and Eq. 8,
  Update DBNet's weights according to the final loss $\mathcal{L}$ defined in Eq. 9.
end for

Output:

The trained DBNet.

where NL means the number of samples in D^t_{train}, $\mathbf{c}_i$ denotes the original prototype of the ith class, $\bar{\mathbf{c}}_i$ implies the mean vector of all features belonging to the ith class, and $\mathbf{c}$ illustrates the center of all samples in D^t_{train}. Through minimizing $\mathcal{L}_{\text{PDM}}$, all samples would be far away from each other, and the samples corresponding to the same class would be close. Consequently, the distances between intra/inter class samples would be large/small so that the discrimination of embedding space could be enhanced.

3) Final Loss Function of CPGL: In sum, the loss function of CPGL is defined as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{PRM}} + \lambda_2 \mathcal{L}_{\text{PDM}} \tag{9}$$

where λ_1 and λ_2 are free parameters that adjust the weights of two regularization terms. Note that, since the units for training are the tasks rather than samples, the $\mathcal{L}_{\text{cls}}$ defined in (3) can be rewritten as

$$L_{\text{cls}} = -\frac{1}{N}\left(\frac{1}{K}\sum_{i}^{N\times K}\left(y_i^s \log\left(\hat{y}_i^s\right)\right) + \frac{1}{Q}\sum_{j}^{N\times Q}\left(y_j^q \log\left(\hat{y}_j^q\right)\right)\right). \tag{10}$$

The specific steps of our CPGL for training DBNet are summarized in Algorithm 1 for clarity.

IV. Experiments

A. Experimental Settings

Four RS scene datasets are selected to test our model and training strategy. The first one is the UC Merced Land-Use (UCM[1]) dataset [20], which contains 2100 RS scenes with the spatial resolution of one foot. Their sizes are 256×256, and they are divided into 21 semantic categories equally. The semantics of each class and the number of scenes within each class are exhibited in Table I. The second one is the aerial image dataset (AID[2]) [75]. AID has 10 000 RS senses with 30 scene classes. Their sizes are 600×600, their pixel resolution ranges from 8 to 1.5 m, and the number of RS scenes in each class varies from 200 to 400. The different semantics and the number of scenes corresponding to each semantic are exhibited in Table II. The third one is a large-scale RS scene dataset named Northwestern Polytechnical University (NWPU)-RESISC45[3] [76]. It contains 31 500 RS scenes with the size of 256×256, and these scenes are grouped into 45 semantic classes. The spatial resolution of RS scenes varies from 30 to 0.2 m. The diverse classes and the number of scenes within each class are exhibited in Table III. The last one is a merged RS scene dataset (RSSDIVCS[4]) [77]. RSSDIVCS has 56 000 scenes, and they are divided into 70 semantic categories equally. The various semantics and the number of scenes per category are exhibited in Table IV. Examples of different datasets can be found on the provided websites. Note that the numbers of RS scenes corresponding to different classes are the same in UCM, NWPU-RESISC45, and RSSDIVCS. For AID, the scene numbers belonging to various classes are imbalanced. We have to admit that our DBNet and the CPGL algorithm do not consider the sample imbalance issue. How to address this problem or reduce its influence is one of our future works.

All of the experiments are implemented by the Pytorch platform using a workstation with NVIDIA GTX TITAN Xp and 128-GB RAM. Here, we select the Adam algorithm as the optimizer. All the models are trained for 2000 iterations. The learning rate is set to be 1×10^{-4} initially, and it drops to one-tenth of the previous every 500 iterations. In DBNet, two ResNets are initialized with the pretrained

TABLE I

Semantics and Numbers of Different Scenes Within UCM. The Numbers of Scenes Within Various Classes Are Equally 100

No.	Semantic	Number	No.	Semantic	Number	No.	Semantic	Number
1	Agricultural	100	8	Forest	100	15	Overpass	100
2	Airplane		9	Freeway		16	Parking Lot	
3	Baseball Diamond		10	Golf Course		17	River	
4	Beach		11	Harbor		18	Runway	
5	Buildings		12	Intersection		19	Sparse Residential	
6	Chaparral		13	Medium Density Residential		20	Storage Tanks	
7	Dense Residential		14	Mobile Home Park		21	Tennis Courts	

TABLE II

Semantics and Numbers of Different Scenes Within AID. The Numbers of Scenes Within Various Semantics Are Different

No.	Semantic	Number	No.	Semantic	Number	No.	Semantic	Number
1	Airport	360	11	Farmland	370	21	Port	380
2	Bare Land	310	12	Forest	250	22	Railway Station	260
3	Baseball Field	220	13	Industrial	390	23	Resort	290
4	Beach	400	14	Meadow	280	24	River	410
5	Bridge	360	15	Medium Residential	290	25	School	300
6	Center	260	16	Mountain	340	26	Sparse Residential	300
7	Church	240	17	Park	350	27	Square	330
8	Commercial	350	18	Parking	390	28	Stadium	290
9	Dense Residential	410	19	Playground	370	29	Storage Tanks	360
10	Desert	300	20	Pond	420	30	Viaduct	420

[1]http://vision.ucmerced.edu/datasets/landuse.html

[2]https://captain-whu.github.io/AID/

[3]https://gcheng-nwpu.github.io/#Datasets

[4]https://github.com/kdy2021/SR-RSKG

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TABLE III

SEMANTICS AND NUMBERS OF DIFFERENT SCENES WITHIN NWPU-RESISC45. THE NUMBERS OF SCENES WITHIN VARIOUS CLASSES ARE EQUALLY 700

No.	Semantic	Number	No.	Semantic	Number	No.	Semantic	Number
1	Airplane	700	16	Golf Course	700	31	Railway Station	700
2	Airport		17	Ground Track Field		32	Rectangular Farmland	
3	Baseball Diamond		18	Harbor		33	River	
4	Basketball Court		19	Industrial Area		34	Roundabout	
5	Beach		20	Intersection		35	Runway	
6	Bridge		21	Island		36	Sea Ice	
7	Chaparral		22	Lake		37	Ship	
8	Church		23	Meadow		38	Snowberg	
9	Circular Farmland		24	Medium Residential		39	Sparse Residential	
10	Cloud		25	Mobile Home Park		40	Stadium	
11	Commercial Area		26	Mountain		41	Storage Tank	
12	Dense Residential		27	Overpass		42	Tennis Court	
13	Desert		28	Palace		43	Terrace	
14	Forest		29	Parking Lot		44	Thermal Power Station	
15	Freeway		30	Railway		45	Wetland	

TABLE IV

SEMANTICS AND NUMBERS OF DIFFERENT SCENES WITHIN RSSDIVCS. THE NUMBERS OF SCENES WITHIN VARIOUS CLASSES ARE EQUALLY 800

No.	Semantic	Number	No.	Semantic	Number	No.	Semantic	Number
1	Airplane	800	25	Grass	800	49	Sclerophyllous Forest	800
2	Airport		26	Harbor		50	Sea	
3	Airport Runway		27	Industrial Area		51	Sea Ice	
4	Bare Land		28	Intersection		52	Sea Island	
5	Baseball Field		29	Lake		53	Ship	
6	Basketball Court		30	Lakeshore		54	Shipping Yard	
7	Beach		31	Mangrove		55	Shrubwood	
8	Bridge		32	Mobile Home Park		56	Snow Mountain	
9	Cemetery		33	Mountain		57	Solar Panel	
10	Christmas Tree Farm		34	Oil Gas Field		58	Sparse Residential	
11	Church		35	Oil Well		59	Square	
12	Circular FarmLand		36	Overpass		60	Stadium	
13	Cloud		37	Palace		61	Storage Tank	
14	Commercial Area		38	Park		62	Swimming Pool	
15	Convention Center		39	Parking Lot		63	Tennis Court	
16	Crosswalk		40	Parking Space Marking		64	Terrace	
17	Dam		41	Pipeline		65	Thermal Power Station	
18	Deadend Road		42	Railway		66	Transformer Station	
19	Dense Residential		43	Railway Station		67	Tree Lined Path	
20	Desert		44	Rectangular Farmland		68	Viaduct Hub	
21	Desert Chaparral		45	Resort		69	Waste Water Plant	
22	Football Field		46	River		70	Wetland	
23	Freeway		47	Roundabout				
24	Golf Course		48	School				

weights. Besides, all RS scenes from different datasets are resized to 224×224 for generating the input pairs. To train DBNet using the CPGL algorithm, we construct three kinds of training tasks for different datasets; they are **Num-5**, **Num-8**, and **Num-10**. In the Num-5 task, we first randomly select five samples from each class to construct the training data. Then, two of them are regarded as the support samples, and three are deemed the query samples. Thus, the Num-5 task is actually N-way-2-shot, where N means the class number of the dataset. In the same way, the Num-8 and Num-10 tasks are N-way-four-shot and N-way-five-shot. Note that all of the remaining samples are used to be the testing data. Several free parameters should be decided in advance; they are k^{top} in PRM, and λ_1 and λ_2 in loss function ((9)). For k^{top}, we set its value to be 2 (Num-5), 2 (Num-8), and 3 (Num-10), respectively. For λ_1 and λ_2, their empirically optimal values for different datasets and different training tasks are summarized in Table V. These values are decided according to the experimental results. Also, their influence on our models is studied in Section IV-D. We have to admit that this manner is inconvenient. Finding an easy and flexible way for free parameters is one of our future works.

TABLE V

EMPIRICALLY OPTIMAL VALUES OF λ_1 AND λ_2 FOR DIFFERENT DATASETS AND DIFFERENT TRAINING TASKS

	UCM		AID		NWPU-RESISC45		RSSDIVCS	
	λ_1	λ_2	λ_1	λ_2	λ_1	λ_2	λ_1	λ_2
Num-5	0.2	1.2	0.6	1.0	0.5	1.0	0.5	1.0
Num-8	0.8	1.6	0.9	1.0	0.5	1.0	0.6	1.0
Num-10	0.5	1.0	0.3	1.0	0.2	1.0	0.3	1.0

To evaluate the proposed method, we adopt overall accuracy (OA) and confusion matrix (CM) as our assessment criteria. OA is defined as the number of correctly classified samples divided by the total number of scenes in the dataset, and it reflects the overall classification performance. CM is a 2-D informative table that is used to analyze the between-class classification errors and confusion degree. Each row of the matrix represents all the image samples of a predicted class, with each column representing the samples of a ground-truth class. Due to the space limitations, the sizes and quality of the exhibited CMs in this manuscript are not optimal. Their high-definition versions can be found in the Supplementary Material.

B. Performance of Our Models

To validate our DBNet and CPGL, we compare their combination (named **CPGL-Net** in the following) with the following eight models, including MSMatch [61], a semisupervised center loss (SSCL) algorithm [36], a pretrained ResNet-18 (Pre-Res), a pretrained EfficientNet-B0 (Pre-Eff) [67], a pretrained Res2Net (Pre-Res2) [68], a discriminative CNN (D-CNN) [58], an Siamese CNN (S-CNN) [78], and an ACNet [32]. The first two models are the semisupervised networks, the middle three models are developed based on the transfer learning framework, and the last three are supervised methods. Note that all methods are accomplished by ourselves, and their free parameters are set according to the original literature. In addition, for the sake of fairness, the experimental settings of the compared methods are the same as the contents discussed in Section IV-A.

1) Results of the UCM Dataset: The performance of different methods counted on the UCM dataset is summarized in Table VI. We can easily find that our CPGL-Net outperforms other models in all cases from observing these results. Taking the Num-8 task as an example, compared with eight models, the improvements achieved by CPGL-Net in OAs are 7.05% (MSMatch), 5.43% (SSCL), 7.01% (Pre-ResNet), 4.87% (Pre-Eff), 4.55% (Pre-Res2), 2.74% (D-CNN), 7.25% (S-CNN), and 3.28% (ACNet), respectively. Another positive inspection is that the standard deviations of CPGL-Net are lower than that of other networks in most cases. This implies that the behavior of CPGL-Net is stable when the volume of training data varies. The reasons for those encouraging results can be summarized as follows. First, due to the random

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TABLE VI
OA VALUES (%) AND STANDARD DEVIATIONS OF DIFFERENT METHODS COUNTED ON FOUR DATASETS

Model	UCM			AID			NWPU-RESISC45			RSSDIVCS		
	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10
MSMatch	79.36 ± 0.77	86.48 ± 1.13	88.43 ± 0.56	66.51 ± 1.07	75.40 ± 0.77	78.87 ± 0.82	59.75 ± 0.86	67.69 ± 0.94	71.45 ± 0.93	70.62 ± 0.72	78.60 ± 0.52	81.30 ± 0.75
SSCL	83.43 ± 0.27	88.10 ± 0.44	90.26 ± 0.68	68.67 ± 1.01	77.11 ± 0.25	80.42 ± 0.66	60.61 ± 0.82	68.41 ± 0.47	71.51 ± 0.61	72.27 ± 0.47	80.59 ± 0.45	82.00 ± 0.40
Pre-Res	81.38 ± 0.72	86.52 ± 0.49	88.69 ± 0.60	73.09 ± 0.48	80.02 ± 0.53	81.36 ± 0.29	63.09 ± 0.89	71.01 ± 0.50	74.51 ± 0.36	76.90 ± 0.55	81.50 ± 0.41	83.51 ± 0.51
Pre-Eff	81.80 ± 0.71	88.66 ± 0.63	90.12 ± 0.36	72.62 ± 0.59	80.45 ± 0.63	82.82 ± 0.86	63.79 ± 0.86	71.53 ± 0.43	76.97 ± 0.31	72.62 ± 0.59	80.45 ± 0.63	82.82 ± 0.86
Pre-Res2	83.56 ± 0.70	88.98 ± 0.69	90.96 ± 0.63	74.53 ± 0.56	81.73 ± 0.42	83.22 ± 0.55	64.42 ± 0.87	72.88 ± 0.42	76.98 ± 0.40	79.13 ± 0.46	82.62 ± 0.36	85.03 ± 0.20
D-CNN	86.28 ± 0.87	90.79 ± 0.64	92.33 ± 0.65	77.28 ± 0.48	82.90 ± 0.19	83.85 ± 0.59	64.37 ± 0.57	73.93 ± 0.80	75.30 ± 0.37	78.93 ± 0.39	83.46 ± 0.31	84.65 ± 0.28
S-CNN	81.83 ± 0.50	86.28 ± 0.52	89.26 ± 0.49	71.57 ± 0.40	78.11 ± 0.74	79.81 ± 0.43	63.15 ± 0.26	70.13 ± 0.60	71.77 ± 0.66	75.39 ± 0.36	80.46 ± 0.25	81.88 ± 0.23
ACNet	84.59 ± 0.78	90.25 ± 1.02	91.24 ± 0.43	74.00 ± 0.71	82.25 ± 0.52	82.74 ± 0.48	65.82 ± 0.75	72.78 ± 0.43	73.36 ± 0.32	77.25 ± 0.59	82.51 ± 0.56	83.97 ± 0.34
CPGL-Net (Ours)	**90.16 ± 0.80**	**93.53 ± 0.29**	**94.59 ± 0.29**	**81.25 ± 0.78**	**85.39 ± 0.09**	**86.03 ± 0.10**	**73.03 ± 0.24**	**78.75 ± 0.29**	**81.19 ± 0.24**	**80.68 ± 0.38**	**85.27 ± 0.09**	**86.71 ± 0.14**

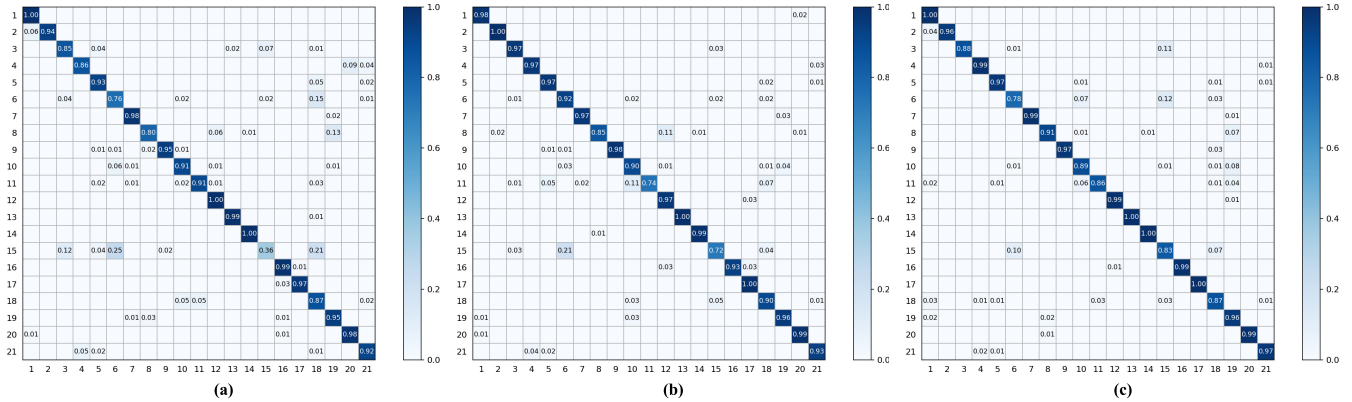


Fig. 3. CMs of CPGL-Net counted on the UCM dataset, which are obtained by the Num-5, Num-8, and Num-10 tasks. (a) CM of Num-5. (b) CM of Num-8. (c) CM of Num-10.

rotation and scaling scene pairs' generation strategies, the proposed DBNet can capture rich content from RS images. Second, by transforming the meta-learning ideas into regularized constraints, our CPGL algorithm can train DBNet with extremely scarce labels. Last but not least, the prototype refining and enhancing methods ensure the effectiveness and robustness CPGL-Net.

To study CPGL-Net's performance in different classes, we draw its CMs (see Fig. 3, and the high-definition version can be found in the Supplementary Material) under the three training tasks. It is easy to find that CPGL-Net can distinguish the scenes from most semantic classes by exploring them. However, its behavior is weak in "Overpass." In addition, most of the scenes corresponding to "Overpass" are misclassified into "Chaparral." By analyzing the scenes in UCM, the reasons are: 1) the scattered objects within "Overpass" (vehicle) and "Chaparral" (trees) have similar outlines due to the spatial resolution; 2) the "Overpass" scenes contain many contents related to "Chaparral"; and 3) the numbers of training data are too small to guide CPGL-Net to differentiate samples from those two classes. Fortunately, with the increase in training data, this deficiency can be improved rapidly.

2) Results of the AID Dataset: The OA values and their standard deviations are exhibited in Table VI. Similar to the results counted on the UCM dataset, the performance of our model is the best among all methods. In addition, the fewer the training samples, the more significant the performance gaps between CPGL-Net and other models. For example, the maximum enhancements in OAs obtained by CPGL-Net under different training tasks are 4.74%1 (Num-5), 9.99% (Num-8), and 7.16% (Num-10). For another example, the OAs of CPGL-Net are 1.11% (Num-5), 1.06% (Num-8), and 1.05% (Num-10) times higher than that of Pre-Res. Meanwhile, to study the performance of different CPGL-Net on each class, their CMs are counted in Fig. 4 (the high-definition version can be found in the Supplementary Material). For most classes, our CPGL-Net performs well. Nevertheless, the behavior of CPGL-Net is weak in "Forest," "Dense Residential," and "Commercial." This is because of the high intrasimilarity between the scenes from them and other classes. Improving our model's performance in this class is one of our future works.

3) Results of the NWPU-RESISC45 Dataset: The OA values and standard deviations of diverse models are also exhibited in Table VI. Like the results of UCM and AID, CPGL-Net is the best among all methods. For the training task Num-5, the enhancements achieved by CPGL-Net are 13.28% (MSMatch), 12.42% (SSCL), 9.94% (Pre-Res), 9.24% (Pre-Eff), 8.61% (Pre-Res2), 8.66% (D-CNN), 9.88% (S-CNN), and 7.21% (ACNet). For the training task Num-8, the improvements obtained by CPGL-Net are 11.06% (MSMatch), 10.34% (SSCL), 7.74% (Pre-Res), 7.22% (Pre-Eff), 5.87% (Pre-Res2), 4.82% (D-CNN), 8.62% (S-CNN), and 5.97% (ACNet). For the training task Num-10, the increases obtained by CPGL-Net

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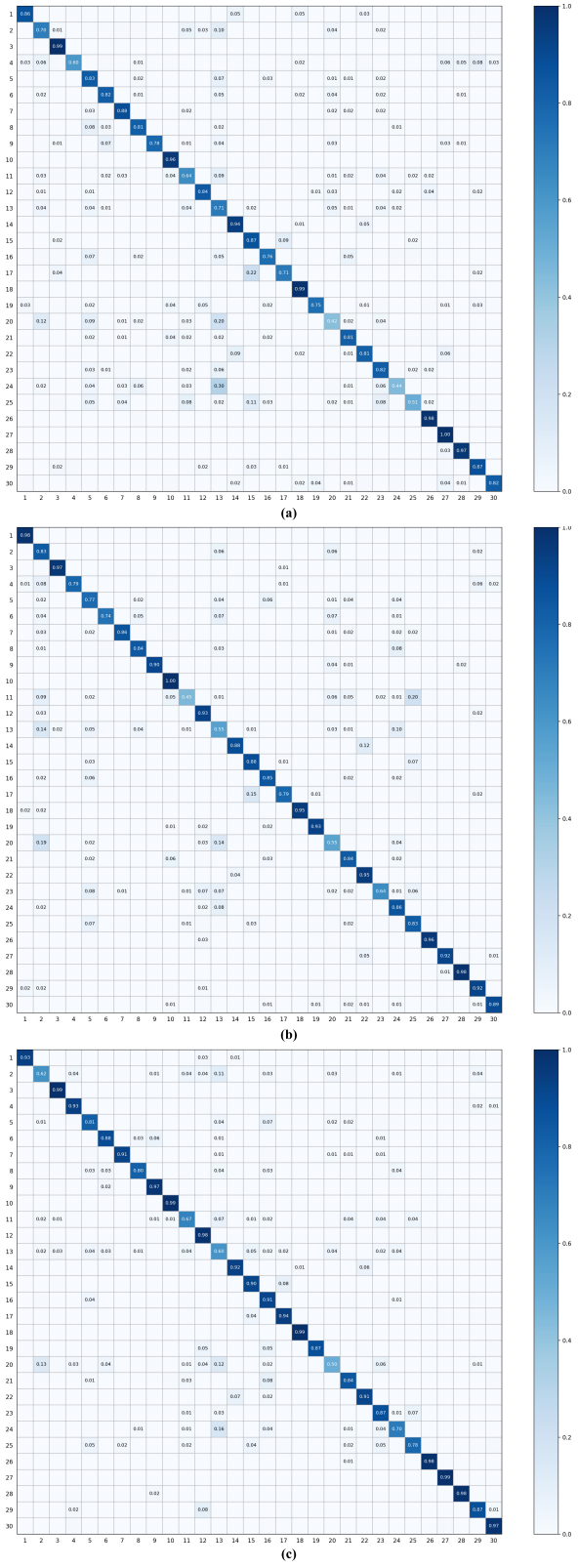


Fig. 4. CMs of CPGL-Net counted on the AID dataset, which are obtained by the Num-5, Num-8, and Num-10 tasks. (a) CM of Num-5. (b) CM of Num-8. (c) CM of Num-10.

are 9.74% (MSMatch), 9.68% (SSCL), 6.68 (Pre-Res)%, 4.22% (Pre-Eff), 4.21% (Pre-Res2), 5.89% (D-CNN), 9.42% (S-CNN), and 7.83% (ACNet). Besides, CMs of different CPGL-Nets counted on the NWPU-RESISC45 dataset are exhibited in Fig. 5 (the high-definition version can be found in the Supplementary Material). It is encouraging that our model works for most semantic classes, such as "Basketball Court," "Bridge," and "Mobile Home Park." However, CPGL-Nets cannot reach what we expect in some classes relatively, including "Commercial Area" and "Sea Ice." This shortcoming can be analyzed from the following aspects. On the one hand, the interclass similarities and intraclass diversities of RS scenes in the NWPU-RESISC45 dataset are high, enhancing classification difficulty. On the other hand, the available prior information is extraordinarily deficient, limiting the model to learn enough discriminative features for categorization tasks. Fortunately, all of the classification results obtained by our model are acceptable, which demonstrates that CPGL-Net is helpful to RSSC.

4) Results of the RSSDIVCS Dataset: The RSSDIVCS dataset is the largest one among the four archives. The OA values and standard deviations counted on RSSDIVCS are exhibited in Table VI. We can still find that our CPGL-Net outperforms other compared methods in all cases. For instance, in the Num-10 task, the enhancements achieved by CPGL-Net in comparison to other methods are 5.41% (MSMatch), 4.71% (SSCL), 3.20% (Pre-Res), 3.98% (Pre-Eff), 1.68% (Pre-Res2), 2.06% (D-CNN), 4.83% (S-CNN), and 2.74% (ACNet). A notable observation is that the performance increases of CPGL-Net are not as high as expected. This is because RSSDIVCS is a merged dataset containing the scenes from UCM, AID, and NWPU-RESISC45. Therefore, the RS scenes' intraclass variance and interclass similarity are higher than that in the other three datasets, which limits the performance of CPGL-Net. At the same time, CMs of various tasks counted on the RSSDIVCS dataset are shown in Fig. 6 (the high-definition version can be found in the Supplementary Material). From observing those figures, it is easy to find that CPGL-Net performs convincingly in most classes. However, due to the distinct interclass similarity, its behavior is weak in some categories, such as "Dam," "Church," "Sea Island," and "Mountain." Reducing the negative impacts of the interclass similarity issue is one of our future works.

C. Effectiveness of CPGL-Net

In this section, we study our CPGL-Net deeply from two aspects. First, we conduct the ablation experiments to observe the contributions of each component within CPGL-Net, including DBNet, PDM, and PRM. Second, we apply the CPGL algorithm to other networks to investigate if it is valuable or not.

To complete the ablation experiments, we construct four networks; they are given as follows.

1) *Net-0:* DBNet.
2) *Net-1:* DBNet + PDM.
3) *Net-2:* DBNet + PRM.
4) *Net-3:* DBNet + PDM + PRM.

Here, Net-0 is our DBNet, and Net-3 is our CPGL-Net. Also, Net-0 is trained by the traditional batch learning strategy, and the CPGL algorithm trains the other three networks. All experimental settings are the same as the contents mentioned in

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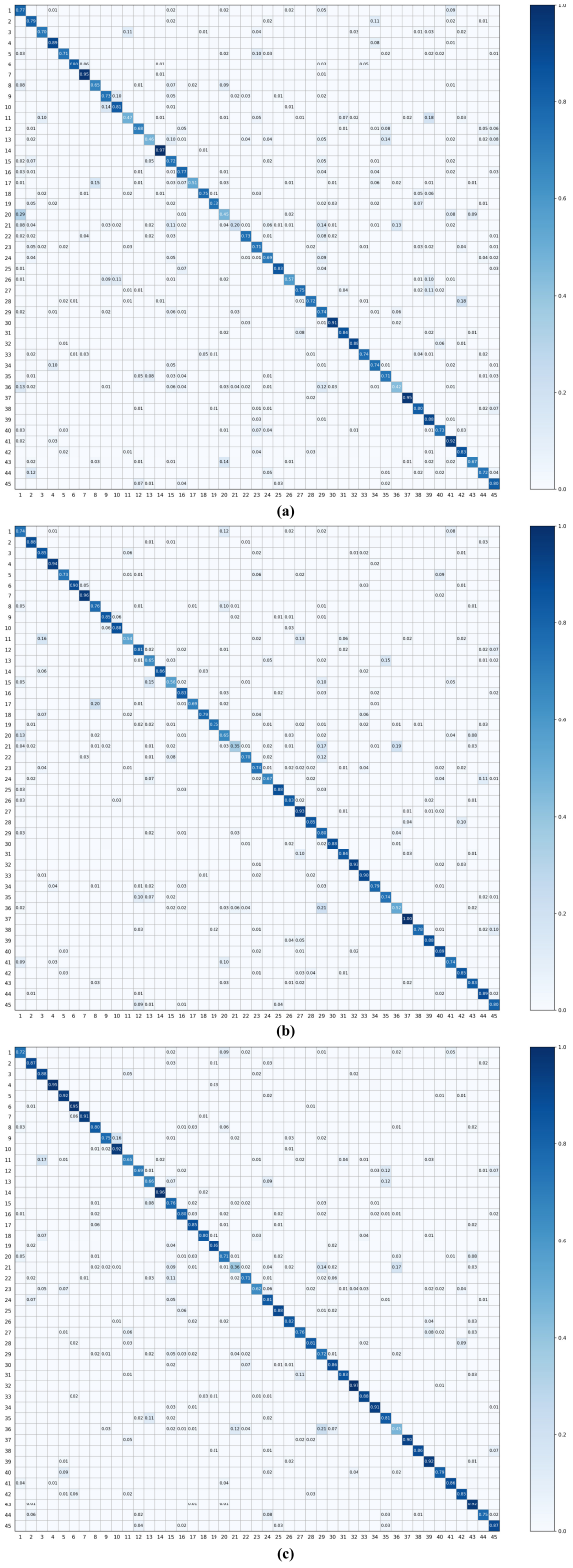


Fig. 5. CMs of CPGL-Net counted on the NWPU-RESISC45 dataset, which are obtained by the Num-5, Num-8, and Num-10 tasks. (a) CM of Num-5. (b) CM of Num-8. (c) CM of Num-10.

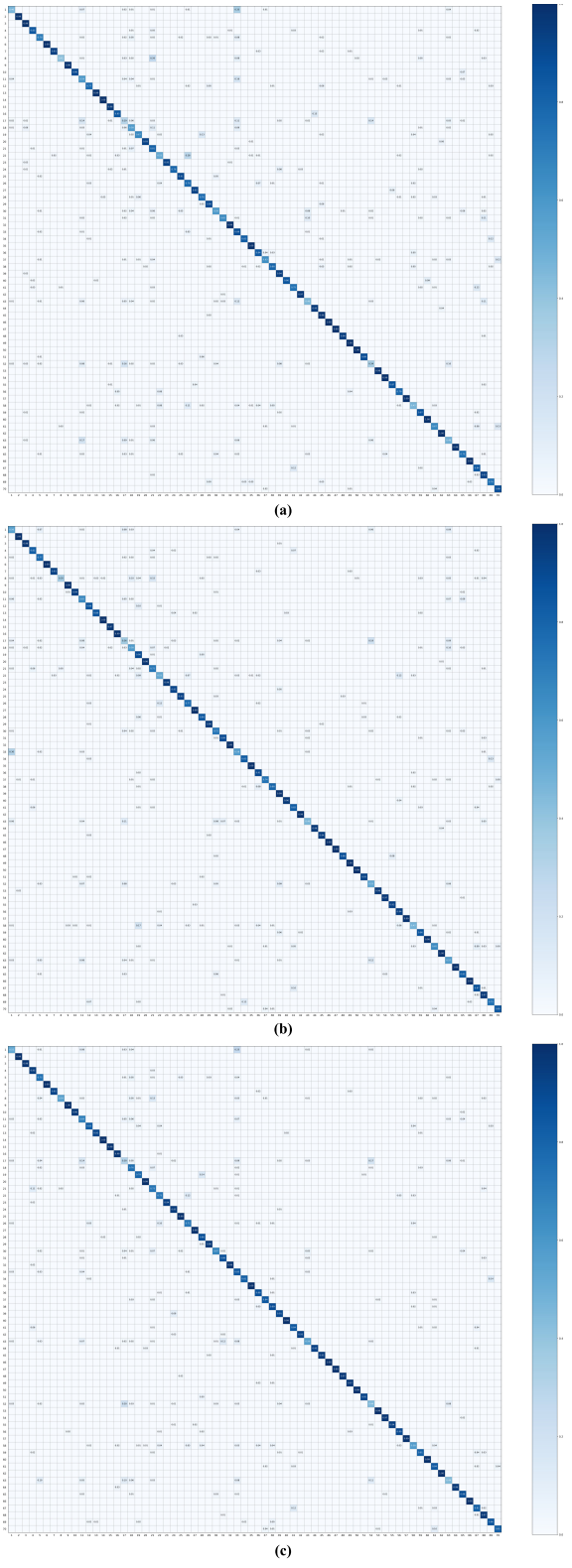


Fig. 6. CMs of CPGL-Net counted on the RSSDIVCS dataset, which are obtained by the Num-5, Num-8, and Num-10 tasks. (a) CM of Num-5. (b) CM of Num-8. (c) CM of Num-10.

Section IV-A. Their performance on different datasets based on various training datasets is displayed in Table VII. We can easily find that the performance of the four networks is incremental, and there is a distinct performance gap between Net-0 and the other three networks. Those observations indicate that: 1) each component of CPGL-Net plays a positive role in ensuring CPGL-Net's behavior and 2) our episode learning algorithm CPGL is suitable for the label lacking scenario.

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TABLE VII

OA VALUES (%) AND STANDARD DEVIATIONS OF NET-0, NET-1, NET-2, AND NET-3 COUNTED ON DIFFERENT DATASETS

Model	UCM			AID			NWPU-RESISC45			RSSDIVCS		
	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10
Net-0	84.42 ± 0.61	89.49 ± 0.22	91.29 ± 0.56	75.80 ± 0.57	82.61 ± 0.27	84.42 ± 0.28	67.45 ± 0.59	74.57 ± 0.31	71.25 ± 0.61	78.51 ± 0.32	83.43 ± 0.40	85.18 ± 0.14
Net-1	88.30 ± 0.48	91.48 ± 0.53	91.48 ± 0.53	77.21 ± 0.73	82.98 ± 0.71	84.72 ± 0.23	69.37 ± 0.56	76.62 ± 0.36	77.49 ± 0.45	80.49 ± 0.18	84.61 ± 0.05	86.34 ± 0.90
Net-2	87.78 ± 0.42	92.16 ± 0.66	92.15 ± 0.64	77.28 ± 0.56	83.28 ± 0.54	85.18 ± 0.50	69.57 ± 0.92	76.98 ± 0.78	78.12 ± 0.42	79.90 ± 0.29	84.25 ± 0.21	86.08 ± 0.24
Net-3	**90.16 ± 0.80**	**93.53 ± 0.29**	**94.59 ± 0.29**	**81.25 ± 0.78**	**85.39 ± 0.09**	**86.03 ± 0.10**	**73.03 ± 0.24**	**78.75 ± 0.29**	**81.19 ± 0.24**	**80.68 ± 0.38**	**85.27 ± 0.09**	**86.71 ± 0.14**

TABLE VIII

OA VALUES (%) AND STANDARD DEVIATIONS OF THREE PRETRAINED MODELS COUNTED ON DIFFERENT DATASETS. ◇ DENOTES THAT THE MODEL IS TRAINED BY THE NORMAL TRAINING STRATEGY. † INDICATES THAT THE MODEL IS TRAINED BY CPGL

Model	UCM			AID			NWPU-RESISC45			RSSDIVCS		
	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10
Pre-Res$^{\diamond}$	81.38 ± 0.72	86.52 ± 0.49	88.69 ± 0.60	73.09 ± 0.48	80.02 ± 0.53	81.36 ± 0.29	63.09 ± 0.89	71.01 ± 0.50	74.51 ± 0.36	76.90 ± 0.55	81.50 ± 0.41	83.51 ± 0.51
Pre-Res†	**86.97 ± 0.61**	**90.10 ± 0.55**	**91.30 ± 0.27**	**76.07 ± 0.60**	**82.06 ± 0.35**	**83.13 ± 0.41**	**66.93 ± 0.47**	**74.33 ± 0.31**	**77.24 ± 0.26**	**78.89 ± 0.37**	**82.45 ± 0.26**	**84.65 ± 0.19**
Pre-Eff$^{\diamond}$	81.80 ± 0.71	88.66 ± 0.63	90.12 ± 0.36	72.62 ± 0.59	80.45 ± 0.63	82.82 ± 0.86	63.79 ± 0.86	71.53 ± 0.43	76.97 ± 0.31	76.61 ± 0.64	81.16 ± 0.43	83.75 ± 0.33
Pre-Eff†	**87.64 ± 0.37**	**92.32 ± 0.33**	**93.51 ± 0.21**	**76.60 ± 0.40**	**84.17 ± 0.09**	**85.45 ± 0.22**	**68.80 ± 0.61**	**75.42 ± 0.33**	**80.58 ± 0.04**	**78.67 ± 0.25**	**83.88 ± 0.09**	**86.31 ± 0.11**
Pre-Res2$^{\diamond}$	83.56 ± 0.70	88.98 ± 0.69	90.96 ± 0.63	74.53 ± 0.56	81.73 ± 0.42	83.22 ± 0.55	64,42 ± 0.87	72.88 ± 0.42	76.98 ± 0.40	79.13 ± 0.46	82.62 ± 0.36	85.03 ± 0.20
Pre-Res2†	**88.30 ± 0.55**	**92.30 ± 0.30**	**93.99 ± 0.13**	**77.70 ± 0.76**	**82.67 ± 0.36**	**85.17 ± 0.13**	**68.92 ± 0.34**	**76.79 ± 0.67**	**80.92 ± 0.18**	**79.96 ± 0.57**	**84.24 ± 0.23**	**86.54 ± 0.38**

TABLE IX

TRAINING TIMES (HOUR) OF DIFFERENT MODELS ON FOUR DATASETS

Model	UCM			AID			NWPU-RESISC45			RSSDIVCS		
	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10	Num-5	Num-8	Num-10
MSMatch	3.5	3.7	3.8	14.4	14.8	15.0	22.6	23.2	23.4	36.4	37.1	37.3
SSCL	4.3	4.5	4.6	17.7	18.2	18.5	28.0	28.7	29.0	45.3	46.1	46.3
Pre-Res	3.3	3.4	3.5	11.7	11.8	12.0	14.7	15.1	15.4	20.6	21.1	21.3
Pre-Eff	3.8	4.0	4.3	13.2	13.8	13.9	27.1	28.4	28.6	41.2	42.4	42.5
Pre-Res2	4.1	4.3	4.6	16.8	16.9	17.1	27.5	27.7	28.0	40.8	41.1	41.4
D-CNN	1.9	2.1	2.2	16.0	16.2	16.3	25.0	25.1	25.2	40.0	40.1	40.2
S-CNN	2.8	2.9	2.9	10.7	10.8	10.9	22.1	23.2	23.3	36.7	37.8	37.9
ACNet	2.3	2.5	2.5	14.9	15.5	15.7	22.3	22.5	22.6	36.9	37.3	37.4
CPGL-Net (Ours)	**0.18**	**0.32**	**0.44**	**0.40**	**0.55**	**0.61**	**0.63**	**0.87**	**0.93**	**0.98**	**1.13**	**1.83**

To further study if the proposed episode training strategy CPGL is effective or not, we apply it to three pretrained models, i.e., Pre-Res, Pre-Eff, and Pre-Res2. The results counted on four datasets are exhibited in Table VIII. Also, three models trained by the regular training strategy are reported for reference. It is easy to find that the models trained by our CPGL algorithm outperform the models obtained by the traditional training strategy in all cases. Taking Pre-Res as an example, when we use the training task Num-5, compared with the normal model (Pre-Res$^{\diamond}$), the enhancements obtained by the CPGL-based model (Pre-Res†) are 5.59% (UCM), 2.98% (AID), 3.84% (NWPU-RESISC45), and 1.99% (RSSDIVCS). The contents discussed above confirm that our CPGL is helpful when the labeled data are deficient.

D. Influence of Free Parameters

As said in Section III, there are two free parameters in our method; they are the hyperparameters λ_1 and λ_2 in the final loss function [see (9)]. In this section, we change their values one by one to study their influence on the proposed CPGL-Net. For simplicity and clarity, the various CPGL-Nets are trained by different Num-8 tasks from four RS scene datasets.

According to the empirically optimal values reported in Section IV-A, we first fix λ_2 to be 1.6, 1.0, 1.0, and 1.0 for the UCM, AID, NWPU-RESISC45, and RSSDIVCS datasets and tune λ_1 from 0.1 to 1.0 with an interval of 0.1. The classification results of different CPGL-Nets counted on four archives are exhibited in Fig. 7. We can find that, apart from the peak values, OAs of the other classification results are similar to each other. Also, all optimal results appear when $\lambda_1 \geq 0.5$ ($\lambda_1 = 0.8$ for UCM, $\lambda_1 = 0.9$ for AID, $\lambda_1 = 0.5$ for NWPU-RESISC45, and $\lambda_1 = 0.6$ for RSSDIVCS). The above observations imply that our method is not sensitive to λ_1, and the bigger the values of λ_1, the better and more stable the performance.

Second, the values of λ_1 are fixed at 0.8, 0.9, 0.5, and 0.6 for the UCM, AID, NWPU-RESISC45, and RSSDIVCS datasets, and λ_2 is varied from 1.0 to 2.0 with an interval

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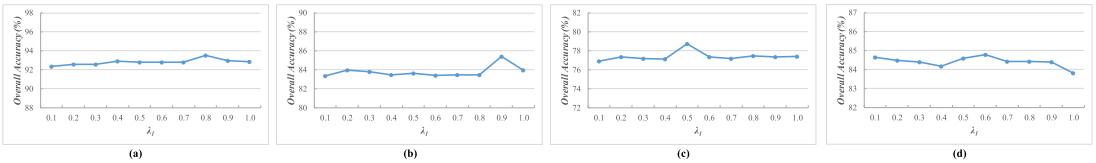


Fig. 7. Influence of λ_1. OAs are counted on the four datasets using the Num-8 tasks. (a) UCM. (b) AID. (c) NWPU-RESISC45. (d) RSSDIVCS.

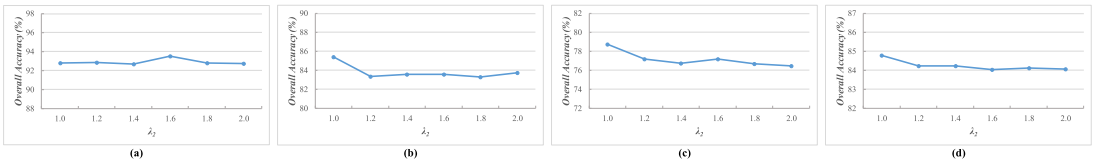


Fig. 8. Influence of λ_2. OAs are counted on the four datasets using the Num-8 tasks. (a) UCM. (b) AID. (c) NWPU-RESISC45. (d) RSSDIVCS.

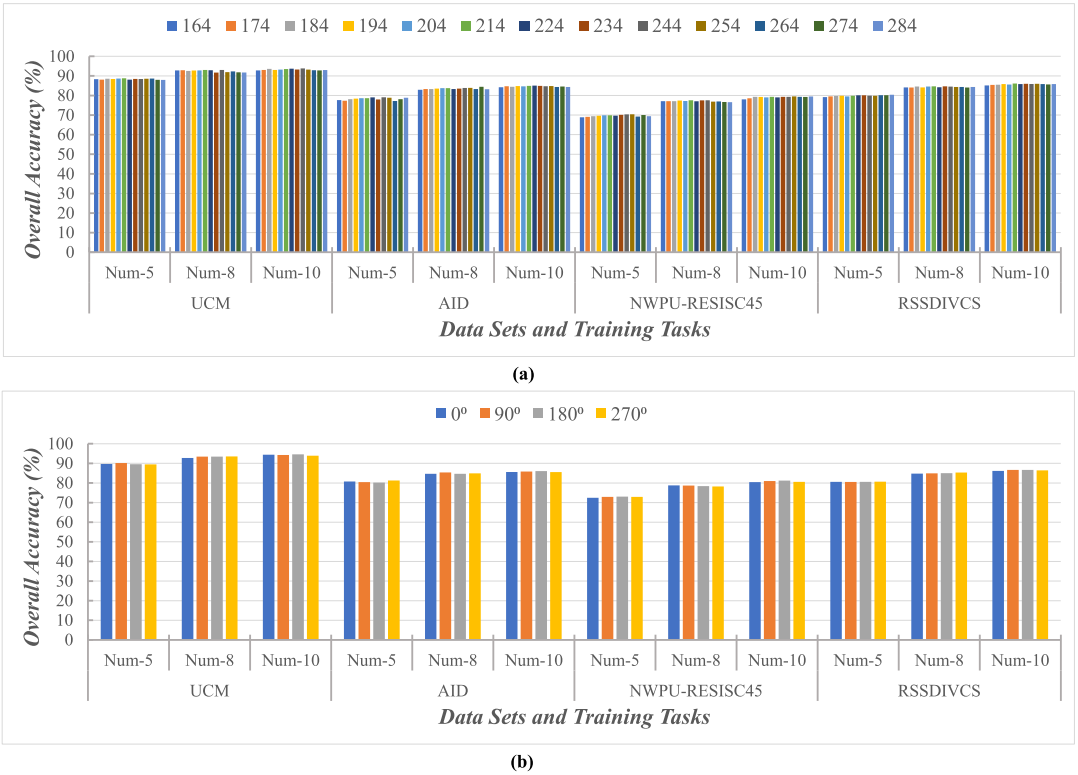


Fig. 9. Influence of scaling sizes and rotation angles. OAs are counted on the four datasets using different training tasks. (a) Scaling sizes. (b) Rotation angles.

of 0.2. The results are displayed in Fig. 8. For UCM, the performance fluctuation of CPGL-Nets with different λ_2's is small, and the peak value appears at $\lambda_2 = 1.6$. For AID, NWPU-RESISC45, and RSSDIVCS, the behavior of CPGL-Nets is getting weaker when λ_2 increases, and the peak values appear at $\lambda_2 = 1.0$. Nevertheless, the gaps between optimal and worst results are narrow. These inspections denote that λ_2 does not impact our method powerfully. Also, its reasonable range is around [1.0, 1.6].

E. Impact of Rotation Angle and Scaling Size

Our DBNet has a dual-branch structure, and its input data are RS scene pairs obtained by the random scaling and rotation. As mentioned in Section III-B, the alternative scaling sizes are $\upsilon \in \{u \pm 10\tau, \tau = 1, 2, 3, 4, 5, 6\}$, where u is the original scene size that equals 224 in this article. Thus, the sizes of the resized RS scenes range from 164×164 to 284×284 with an interval of 10×10. Meanwhile, the available rotation angles are $\{0°, 90°, 180°, 270°\}$. We do the following experiments to ensure suitable scaling sizes and rotation angles for the inference stage. First, we fix a scaling size and rotate original scenes randomly to generate scene pairs. Second, we arbitrarily resize original scenes to an alternative scaling size and fix a rotation angle to produce input pairs. Based on those input data, we can get various CPGL-Nets, and their classification performance on different datasets is illustrated in Fig. 9. From analyzing these results, we can find that different scaling sizes and rotation angles do not fluctuate our model

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significantly. In other words, when our models are trained, the scaling size and rotation angle are not crucial for the final classification results.

F. Time Costs

To study the efficiency of our CPGL-Net, we count its training time for different datasets and various tasks. Also, the compared methods' training times are counted for reference. The results are summarized in Table IX. By observing the table, we can find that CPGL-Net is time-saving compared to the other methods. This owes to the episode training manner and the proper loss function, ensuring that the network can be trained easily. Interestingly, the training cost increment of CPGL-Net is not substantial when the scale of the dataset increases. This implies that our model can be used for larger datasets. These encouraging results illustrate the superiority of the proposed CPGL-Net.

V. Conclusion

This article proposes a DBNet and a specific training method (CPGL) to address RSSC tasks with scarce labels. First, considering the characteristics of RS images, DBNet mines multiscale and multiangle information hidden in RS images. Thus, the rich land covers with various sizes, angles, and outlines can be fully explored. Second, since a few samples have semantic labels, CPGL formulates the training process in a metaway. In detail, rather than using a single sample, CPGL first selects the sample tasks to be the training units. Then, two prototypes refining and enhancement strategies are developed to emphasize the intraclass and interclass relationships between limited samples. Finally, a joint loss function is defined to update the DBNet's parameters. The positive results counted on the four RS datasets scene benchmarks demonstrate that our model and training method are helpful to RSSC tasks when the labeled samples are insufficient.

Although the proposed DBNet and the CPGL algorithm achieve positive results on different RS datasets, there is still room for improvement. First, the proposed models do not consider the sample imbalance problem, limiting their practical application performance. Second, it is hard and time-consuming to find the proper hyperparameters for our models, especially when the scale of the RS dataset is large. Addressing the limitations mentioned above is our future work. Besides, developing the models based on DBNet and CPGL to deal with the unseen sample problem is another important work.

References

[1] X. Yao, J. Han, G. Cheng, X. Qian, and L. Guo, "Semantic annotation of high-resolution satellite images via weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3660–3671, Jun. 2016.

[2] X. Tang *et al.*, "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[3] R. Dong, D. Xu, J. Zhao, L. Jiao, and J. An, "Sig-NMS-based faster R-CNN combining transfer learning for small target detection in VHR optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8534–8545, Nov. 2019.

[4] Y. Yang *et al.*, "AR^{2}Det: An accurate and real-time rotational one-stage ship detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5605414.

[5] R. Dong, L. Jiao, Y. Zhang, J. Zhao, and W. Shen, "A multi-scale spatial attention region proposal network for high-resolution optical remote sensing imagery," *Remote Sens.*, vol. 13, no. 17, p. 3362, 2021.

[6] Y. Li *et al.*, "Recurrent attention and semantic gate for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[7] X. Zhang, X. Wang, X. Tang, H. Zhou, and C. Li, "Description generation for remote sensing images using attribute attention mechanism," *Remote Sens.*, vol. 11, no. 6, p. 612, Mar. 2019.

[8] X. Tang, L. Jiao, W. J. Emery, F. Liu, and D. Zhang, "Two-stage reranking for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5798–5817, Oct. 2017.

[9] C. Liu, J. Ma, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Deep hash learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3420–3443, Apr. 2021.

[10] X. Tang *et al.*, "Meta-hashing for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2021.

[11] U. Maulik and D. Chakraborty, "Remote sensing image classification: A survey of support-vector-machine-based advanced techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 33–52, Mar. 2017.

[12] Y. Li, H. Zhang, X. Xue, Y. Jiang, and Q. Shen, "Deep learning for remote sensing image classification: A survey," *Wiley Interdiscipl. Rev., Data Mining Knowl. Discovery*, vol. 8, no. 6, p. e1264, 2018.

[13] R. Dong, D. Xu, L. Jiao, J. Zhao, and J. An, "A fast deep perception network for remote sensing scene classification," *Remote Sens.*, vol. 12, no. 4, p. 729, 2020.

[14] X. Huang, X. Han, S. Ma, T. Lin, and J. Gong, "Monitoring ecosystem service change in the city of Shenzhen by the use of high-resolution remotely sensed imagery and deep learning," *Land Degradation Develop.*, vol. 30, no. 12, pp. 1490–1501, Jul. 2019.

[15] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[16] Y. Gu, Y. Wang, and Y. Li, "A survey on deep learning-driven remote sensing image scene understanding: Scene classification, scene retrieval and scene-guided object detection," *Appl. Sci.*, vol. 9, no. 10, p. 2110, May 2019.

[17] L. Jiao, X. Tang, B. Hou, and S. Wang, "SAR images retrieval based on semantic classification and region-based similarity measure for Earth observation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 8, pp. 3876–3891, Aug. 2015.

[18] X. Tang and L. Jiao, "Fusion similarity-based reranking for SAR image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 2, pp. 242–246, Feb. 2017.

[19] X. Tang, L. Jiao, and W. J. Emery, "SAR image content retrieval based on fuzzy similarity and relevance feedback," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1824–1842, May 2017.

[20] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, 2010, pp. 270–279.

[21] H. Jegou, M. Douze, C. Schmid, and P. Pérez, "Aggregating local descriptors into a compact image representation," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 3304–3311.

[22] F. Perronnin, J. Sánchez, and T. Mensink, "Improving the Fisher kernel for large-scale image classification," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2010, pp. 143–156.

[23] X. Bian, C. Chen, L. Tian, and Q. Du, "Fusing local and global features for high-resolution scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 2889–2901, Jun. 2017.

[24] A. Stumpf and N. Kerle, "Object-oriented mapping of landslides using random forests," *Remote Sens. Environ.*, vol. 115, no. 10, pp. 2564–2577, Oct. 2011.

[25] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[26] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1097–1105.

[27] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

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[29] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[30] L. Fu, D. Zhang, and Q. Ye, "Recurrent thrifty attention network for remote sensing scene recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8257–8268, Oct. 2021.

[31] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.

[32] X. Tang, Q. Ma, X. Zhang, F. Liu, J. Ma, and L. Jiao, "Attention consistent network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2030–2045, 2021.

[33] J. Ma, M. Li, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Homo–heterogenous transformer learning framework for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2223–2239, 2022.

[34] X. Yu, X. Wu, C. Luo, and P. Ren, "Deep learning in remote sensing scene classification: A data augmentation enhanced convolutional neural network framework," *GISci. Remote Sens.*, vol. 54, no. 5, pp. 741–758, 2017.

[35] D. Ma, P. Tang, and L. Zhao, "SiftingGAN: Generating and sifting labeled samples to improve the remote sensing image scene classification baseline *in vitro*," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1046–1050, Jul. 2019.

[36] J. Zhang, M. Zhang, B. Pan, and Z. Shi, "Semisupervised center loss for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1362–1373, 2020.

[37] K. Sohn *et al.*, "FixMatch: Simplifying semi-supervised learning with consistency and confidence," 2020, *arXiv:2001.07685*.

[38] M. Assran *et al.*, "Semi-supervised learning of visual features by non-parametrically predicting view assignments with support samples," 2021, *arXiv:2104.13963*.

[39] X. Yang, Z. Song, I. King, and Z. Xu, "A survey on deep semi-supervised learning," 2021, *arXiv:2103.00550*.

[40] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, pp. 1–34, 2020.

[41] S. Hochreiter, A. S. Younger, and P. R. Conwell, "Learning to learn using gradient descent," in *Proc. Int. Conf. Artif. Neural Netw.* Berlin, Germany: Springer, 2001, pp. 87–94.

[42] A. A. Rusu *et al.*, "Meta-learning with latent embedding optimization," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–17. [Online]. Available: https://openreview.net/forum?id=BJgklhAcK7

[43] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.

[44] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3630–3638.

[45] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[46] R. P. de Lima and K. Marfurt, "Convolutional neural network for remote-sensing scene classification: Transfer learning analysis," *Remote Sens.*, vol. 12, no. 1, p. 86, 2020.

[47] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[48] X. Li, Z. Sun, J.-H. Xue, and Z. Ma, "A concise review of recent few-shot meta-learning methods," *Neurocomputing*, vol. 456, pp. 463–468, Oct. 2021.

[49] N. Bendre, H. T. Marín, and P. Najafirad, "Learning from few samples: A survey," 2020, *arXiv:2007.15484*.

[50] W. Luo, Y. Li, R. Urtasun, and R. Zemel, "Understanding the effective receptive field in deep convolutional neural networks," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 4905–4913.

[51] N. He, L. Fang, S. Li, A. Plaza, and J. Plaza, "Remote sensing scene classification using multilayer stacked covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6899–6910, Dec. 2018.

[52] X. Lu, H. Sun, and X. Zheng, "A feature aggregation convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7894–7906, Oct. 2019.

[53] X. Wang, S. Wang, C. Ning, and H. Zhou, "Enhanced feature pyramid network with deep semantic embedding for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7918–7932, Sep. 2021.

[54] S. Chaudhari, V. Mithal, G. Polatkan, and R. Ramanath, "An attentive survey of attention models," 2019, *arXiv:1904.02874*.

[55] M. Sundermeyer, R. Schlüter, and H. Ney, "LSTM neural networks for language modeling," in *Proc. 13th Annu. Conf. Int. Speech Commun. Assoc.*, Sep. 2012, pp. 1–14.

[56] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[57] Y. Guo, J. Ji, X. Lu, H. Huo, T. Fang, and D. Li, "Global-local attention network for aerial scene classification," *IEEE Access*, vol. 7, pp. 67200–67212, 2019.

[58] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[59] S. Chopra, R. Hadsell, and Y. LeCun, "Learning a similarity metric discriminatively, with application to face verification," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 539–546.

[60] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2016, pp. 499–515.

[61] P. Gómez and G. Meoni, "MSMatch: Semi-supervised multispectral scene classification with few labels," 2021, *arXiv:2103.10368*.

[62] W. Han, R. Feng, L. Wang, and Y. Cheng, "A semi-supervised generative framework with deep learning features for high-resolution remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 23–43, Nov. 2018.

[63] J. Kang, R. Fernandez-Beltran, Z. Ye, X. Tong, P. Ghamisi, and A. Plaza, "High-rankness regularized semi-supervised deep metric learning for remote sensing imagery," *Remote Sens.*, vol. 12, no. 16, p. 2603, 2020.

[64] F. Zhuang *et al.*, "A comprehensive survey on transfer learning," *Proc. IEEE*, vol. 109, no. 1, pp. 43–76, Jan. 2021.

[65] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[66] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.

[67] D. Zhang, Z. Liu, and X. Shi, "Transfer learning on EfficientNet for remote sensing image classification," in *Proc. 5th Int. Conf. Mech., Control Comput. Eng. (ICMCCE)*, Dec. 2020, pp. 2255–2258.

[68] A. Das and S. Chandran, "Transfer learning with Res2Net for remote sensing scene classification," in *Proc. 11th Int. Conf. Cloud Comput., Data Sci. Eng. (Confluence)*, Jan. 2021, pp. 796–801.

[69] Y. Liang, S. T. Monteiro, and E. S. Saber, "Transfer learning for high resolution aerial image classification," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, Oct. 2016, pp. 1–8.

[70] H. Li *et al.*, "RS-MetaNet: Deep meta metric learning for few-shot remote sensing scene classification," 2020, *arXiv:2009.13364*.

[71] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7844–7853, Sep. 2021.

[72] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[73] Y. Liu, L. Zhang, Z. Han, and C. Chen, "Integrating knowledge distillation with learning to rank for few-shot scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[74] Y. Liu, Y. Zhong, and Q. Qin, "Scene classification based on multiscale convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7109–7121, Dec. 2018.

[75] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[76] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[77] Y. Li, D. Kong, Y. Zhang, Y. Tan, and L. Chen, "Robust deep alignment network with remote sensing knowledge graph for zero-shot and generalized zero-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 145–158, Sep. 2021.

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[78] X. Liu, Y. Zhou, J. Zhao, R. Yao, B. Liu, and Y. Zheng, "Siamese convolutional neural networks for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1200–1204, Aug. 2019.



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